

Phishing Email Detection Using Large Language Models

A Literature Review (2023–2026)

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ID	Name
011221002	Md. Jakaria Alam Saimon
011221376	Jahid Ibna Sinha
011221372	Nowshin Anjum Faria

1 Introduction

Phishing is still one of the most common ways attackers get initial access to accounts and organizations. Most phishing attacks start with an email that pretends to come from a trusted brand or person and tries to make the victim click a link or share credentials. Traditional defenses like blocklists and rule based spam filters struggle here because attackers keep changing their wording, and now they can even use generative AI to write convincing emails at almost no cost. Because of this, a lot of recent research between 2023 and 2026 has looked at whether large language models (LLMs) themselves can be used on the defense side, to read an email the way a careful human would and decide if it is phishing or not.

In this review we study five recent papers on this topic. For each paper we followed the reading order suggested in class: the abstract for the overall summary, the second last paragraph of the introduction for the motivation, then the dataset, the methodology, and finally the conclusion and future work sections to understand the limitations. At the end we compare all five papers in a table and point out the gap we want to work on. One practical reason we chose this topic is that the datasets used in these papers (Nazario, Enron, SpamAssassin, CEAS_08, and Kaggle phishing email corpora) are all publicly available text datasets, so the experiments can be reproduced on a normal computer with open models accessed through cheap APIs.

2 Review of Selected Papers

2.1 Heiding et al., “Devising and Detecting Phishing Emails Using Large Language Models” (IEEE Access, 2024)

This paper looks at both sides of the problem: using LLMs to create phishing emails and using them to detect phishing. The authors ran a human experiment with 112 university participants who received four types of emails: generic control phishing, emails generated by GPT-4, emails manually written using the V-Triad (a set of social engineering design rules), and a combination of V-Triad and GPT-4. In the introduction the authors motivate the work by pointing out that manual methods like the V-Triad work well but are time consuming and need real social engineering knowledge, while an attacker with an LLM chatbot can now produce a good phishing email from a few collected data points in seconds. There is no standard public dataset here; the data comes from the recruited participants and the emails created for the study (click rates were 19–28% for control, 30–44% for GPT-4, 69–79% for V-Triad, and 43–81% for the combined version). For detection, they gave GPT-4, Claude, Bard, and LLaMA a set of emails from each category plus legitimate marketing emails and asked each model about the intent of the email, how suspicious it is, what the user should do, and whether it looked AI written. The models often performed close to humans and sometimes better at spotting malicious intent. The main limitations the authors admit are the small sample size (112 people), the fact that the sender address was a spoofed personal Gmail account, and possible information leakage between participants. As future work they mention automating the full attack pipeline and building personalized security training using LLMs.

2.2 Koide et al., “ChatSpamDetector: Leveraging Large Language Models for Effective Phishing Email Detection” (2024)

The abstract presents ChatSpamDetector, a system that converts a raw email into a prompt for an LLM and gets back not just a phishing verdict but also detailed reasoning that a user can read. In the second last part of the introduction the authors say that earlier work mostly studied how LLMs can generate phishing, while the ability of LLMs to distinguish real emails from fake ones had not been properly demonstrated, and they claim their system is the first proper evaluation of that ability. For the dataset, they built a balanced corpus of 1,010 phishing emails from the public phishing_pot repository (collected between August 2022 and

October 2023, covering 19 languages and 193 brands) and 1,000 legitimate emails sampled from the CSDMC spam corpus. The methodology is a four stage pipeline: parse the .eml file, simplify the body so it fits in the token limit, build a chain of thought prompt with role assignment, and parse the JSON answer which contains the verdict, a confidence score, the impersonated brand, and the reasons. With GPT-4 the system reached 99.70% accuracy, clearly above GPT-3.5 (97.03%), Llama 2 (94.06%), Gemini Pro (98.12%), and conventional baseline filters. The conclusion admits some real limitations: the system only handles emails that contain URLs, so attachment based malware emails and plain money transfer scams are out of scope, and the LLM cannot recognize brands or domains that appeared after its training cutoff. Their suggested future direction is adding retrieval augmented generation so the model can check current domain information.

2.3 Uddin, Mahiuddin and Sarker, “An Explainable Transformer-based Model for Phishing Email Detection” (2024)

This paper takes the fine tuning route instead of prompting a big commercial model. The abstract describes a fine tuned RoBERTa transformer for phishing email classification, with a strong focus on explainability. In the introduction the authors motivate this by saying that BERT style models perform very well but act as black boxes, and in cybersecurity a prediction that cannot be explained is hard to trust, so explainable AI should be a core part of the detector, not an afterthought. The dataset is the public Phishing Email Dataset from Kaggle with 18,650 emails (11,322 safe and 7,328 phishing). Since the classes are imbalanced, they used cleaning plus synonym replacement augmentation to balance it at 11,322 emails per class, then split it 64/16/20 for training, validation, and testing. The methodology fine tunes RoBERTa-base (125M parameters) for 5 epochs with the AdamW optimizer, and then applies two explanation techniques, LIME and Transformers Interpret. Because the two methods sometimes disagree about which words mattered, the authors propose LITA, a hybrid that combines the normalized scores of both. The model reaches 98.45% test accuracy. In the conclusion they accept that the model was evaluated only on phishing emails (spam was deliberately excluded) and that some earlier works using mixed phishing plus spam datasets report higher numbers. Future work includes trying decoder LLMs such as LLaMA, Gemini, Falcon, and Mistral, and moving to multilingual datasets.

2.4 Lin, Liu and Fan, “Improving Phishing Email Detection Performance of Small Large Language Models” (2025)

This paper asks a very practical question: can small LLMs of around 3 billion parameters, which run on a consumer GPU, detect phishing as well as the big models. The abstract reports that with the right techniques the answer is close to yes. The motivation given in the introduction is that using such small models directly gives poor results, for example Qwen2.5-1.5B-Instruct got only 0.388 accuracy on SpamAssassin and LLaMA-3.2-3B only 0.587, so the gap to standard sized LLMs has to be closed somehow. The experiments use the public SpamAssassin and CEAS_08 email corpora, with a training subset of 1,000 samples used for building explanations. The methodology has three parts: prompt engineering that forces the model to write its reasoning before the final label, explanation augmented fine tuning where GPT-4o-mini generates an explanation for each training email and the small models (LLaMA-3.2-3B-Instruct, Phi-4-mini, Qwen2.5-1.5B-Instruct) are fine tuned on these with LoRA, and finally model ensembling using token probability based confidence and majority voting. The fine tuned small models improved a lot in accuracy and F1 and generalized well to unseen phishing datasets. The stated limitations are that the evaluation covers only a small number of datasets, transferability across very different email domains is not fully studied, and there is no proper cost benefit analysis, which they leave as future work.

2.5 Kuikel, Piplai and Aggarwal, “Evaluating Large Language Models for Phishing Detection, Self-Consistency, Faithfulness, and Explainability” (2025)

This last paper does not propose a new detector; it asks whether the explanations LLMs give for their phishing verdicts can actually be trusted. The abstract explains that the authors fine tuned several transformer models and then measured whether each model’s explanation really matches the tokens that drove its prediction, using a metric called CC-SHAP. The introduction motivates this by noting that even models that perform well often fail to generalize to evolving attack strategies and lack interpretability, which is a problem when an analyst has to act on the model’s output. The dataset is a balanced set of 5,000 emails: 2,500 phishing emails from the Nazario corpus (a 2015 to 2024 subset) and 2,500 legitimate emails from the Enron corpus. The methodology compares BERT, Llama-2-7B, Llama-3-8B, and Wizard-7B fine tuned in three different ways: plain binary sequence classification, contrastive learning, and direct preference optimization (DPO). The interesting result is that accuracy and faithfulness do not move together: the Llama models had strong prediction to explanation

alignment but lower accuracy, while Wizard was more accurate but less self consistent. In the conclusion the authors raise an open question about whether these models should imitate human style uncertainty or be strictly deterministic, and they call for human centric fine tuning and better evaluation metrics as future work, while noting the study covers a limited set of models and a single dataset pair.

3 Comparison of the Five Papers

Paper	Year	Approach / Models	Dataset	Key Result	Main Limitation
Heiding et al. [1]	2024	Human study + prompting GPT-4, Claude, Bard, LLaMA for detection	Own study emails, 112 participants (no public corpus)	LLMs matched and sometimes beat humans at spotting phishing intent	Small sample, spoofed Gmail sender, no standard dataset
Koide et al. [2]	2024	Prompt pipeline (chain of thought, JSON output) on GPT-4 and others	1,010 phishing (phishing_pot) + 1,000 ham (CSDMC)	99.70% accuracy with GPT-4	Only URL based phishing, model knowledge cutoff for new brands
Uddin et al. [3]	2024	Fine tuned RoBERTa + LIME, Transformers Interpret, LITA hybrid XAI	Kaggle Phishing Email Dataset, 18,650 emails (balanced to 22,644)	98.45% accuracy with word level explanations	Phishing only (no spam), single English dataset
Lin et al. [4]	2025	Small LLMs (1.5B–3B) + LoRA fine tuning on GPT-4o-mini explanations + ensemble	SpamAssassin, CEAS_08	Small models approach standard LLM performance	Few datasets, no cost benefit analysis
Kuikel et al. [5]	2025	BERT, Llama-2/3, Wizard fine tuned (classification, contrastive, DPO), CC-SHAP faithfulness	Nazario (2,500 phish) + Enron (2,500 ham)	Accuracy and explanation faithfulness do not correlate	Limited models and datasets, open evaluation questions

4 Research Gap and Our Direction

Looking at the five papers together, a clear pattern shows up. The best raw accuracy comes from prompting large closed models like GPT-4 [2], but that is expensive at scale and the explanations are not guaranteed to be faithful [5]. Fine tuned encoder models like RoBERTa are cheap and accurate [3] but were tested on a single English dataset, and small open LLMs can get close to the big ones but only with help from an expensive teacher model [4]. Almost nobody evaluates across several public datasets at once, and cost is rarely reported even though it decides what is actually deployable.

Our plan is to work exactly in that gap: benchmark a few small open weight LLMs (accessed through OpenRouter) against a fine tuned baseline on multiple free public corpora (Nazario, Enron, SpamAssassin, and the Kaggle phishing set), and report accuracy together with the actual API cost per thousand emails. All the datasets are free and small enough for a normal laptop, and based on current OpenRouter prices the full set of experiments should stay under a five dollar budget.

References

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